

END-Semester Project Report

Evaluating Physics-Informed Neural Networks as a Framework for Scientific Computing

A project report submitted in fulfilment of
the requirements for the Award of the Degree of

Bachelor of Technology in Engineering Physics

by

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May 2026

CERTIFICATE

This is to certify that the dissertation entitled “**Evaluating Physics-Informed Neural Networks as a Framework for Scientific Computing**” submitted by **Shonjith Shaji**, to the National Institute of Technology, Calicut, towards fulfilment of the requirements for the Award of the Degree of **Bachelor of Technology in Engineering Physics** is a bona fide record of the work carried out by him under my supervision and guidance.

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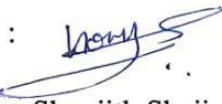
I would also like to express my heartfelt gratitude to my faculty advisor, Dr. Natesan Yogesh, for his continuous support, guidance, and constant encouragement throughout this project. His willingness to help at any stage made the entire journey much easier.

Declaration

I hereby declare that the work presented in this Phase-II end semester project report entitled "*Evaluating Physics-Informed Neural Networks as a Framework for Scientific Computing*" is an original work carried out by me under the guidance of the project supervisor, except where otherwise acknowledged through proper references and citations.

I further declare that this report has not been submitted, either wholly or partially, for the award of any degree, diploma, or other academic qualification at this or any other institution. The results, discussions, and interpretations presented in this report are true to the best of my knowledge and conform to the academic integrity standards of the institute.

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Abstract

This report evaluates the viability of Physics-Informed Neural Networks (PINNs) as an alternative or complementary framework to classical numerical methods, specifically the Explicit Finite Difference Method (FDM), for scientific computing. The study conducts a rigorous comparative analysis across a range of physical systems, including the one-dimensional Non-linear Schrödinger Equation and one- and two-dimensional diffusion equations under various symmetric and asymmetric boundary conditions. Furthermore, the report explores the unique capabilities of PINNs in solving an inverse problem to discover a hidden physical defect purely from noisy sensor data.

The findings demonstrate that FDM remains highly computationally efficient and accurate for standard forward problems with simple, well-defined boundaries. However, PINNs offer distinct advantages in handling complex boundaries without the constraints of spatial discretization and serve as a transformative tool for parameter discovery in inverse problems. The research concludes that deep learning frameworks and classical numerical methods should be utilized as complementary tools, chosen based on the specific geometric, parametric, and computational complexities of the physical system in question.

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1 Problem Statement

The primary aim of this project is to implement, analyze, and evaluate Physics-Informed Neural Networks as a framework for modeling physical systems governed by partial differential equations, and to understand their strengths and limitations in practical scientific computing applications. The problem also involves the analysis of whether Physics-Informed Neural Networks can be a replacement of traditional numerical methods, which is widely accepted for scientific computing.

Introduction

Recent advances in deep learning have introduced Physics-Informed Neural Networks (PINNs) as a novel framework that integrates physical laws directly into neural network training using automatic differentiation. Unlike conventional neural networks that rely purely on data, PINNs embed governing differential equations into the loss function, which helps it to maintain the physics behind the problem enabling data-efficient and mesh-free modeling of physical systems.

While the theoretical base of PINNs has been already established its practicality in scientific computing is still under evaluation. The understanding of how effectively PINNs can enforce Physical constraints and its comparison with traditional computational methods are needed to be done before generalising PINNs into a complete scientific computing framework.

The comparison is done by implementing PINNs and Explicit finite difference method framework to a variety of physics problems and then the performance are compared in terms of stability, accuracy, computational efficiency, and physical consistency.

2 Objectives of the Project

The ultimate aim of this project is to evaluate whether PINNs is an alternative tool to the classical computational PDE solvers. This aim is been evaluated by following the given objectives:

1. Solve Forward Problems

- (a) Implement PINN and discrete Explicit Finite Difference Method (FDM) solvers for the 1D diffusion equation across different physical scenarios, which includes symmetric (two absorbing) and asymmetric (one absorbing, one reflecting) boundary conditions.
- (b) Extend both frameworks to a higher dimensional spaces by solving the 2D bounded diffusion equation.
- (c) Implement PINN to solve the 1D Nonlinear Schrödinger Equation.

2. Solve Inverse Problems

- Create a dual-network PINN architecture to solve an inverse problem by discovering a hidden physical defect purely from noisy sensor data without prior knowledge of the underlying parameter distribution.

3. Conduct a Comparative Analysis

- Evaluate the performance of PINNs against FDM in terms of:
 - (a) Accuracy
 - (b) Stability
 - (c) Computational Efficiency
 - (d) Flexibility in Handling Boundarie

3 Literature Review

Physics-Informed Neural Networks (PINNs), introduced by Raissi, Perdikaris, and Karniadakis (2019) [1], provide a deep learning framework that integrates physical laws directly into neural network training. Unlike conventional data-driven models that approximate mappings purely from data, PINNs incorporate governing partial differential equations (PDEs) into the loss function using automatic differentiation. This enables the network to satisfy both observed data and the underlying physical laws simultaneously.

The framework considers general nonlinear PDEs of the form:

$$u_t + \mathcal{N}[u; \lambda] = 0 \quad (1)$$

where $u(t, x)$ represents the solution and $\mathcal{N}$ is a nonlinear differential operator parameterized by λ [1].

The loss function is composed of a data loss, enforcing initial and boundary conditions, and a physics loss, minimizing the PDE residual at selected collocation points. This structure regularizes the learning process and improves performance in small-data regimes.

The authors propose two main formulations:

Continuous-Time PINNs — The neural network directly approximates $u(t, x)$, and derivatives required in the PDE are computed via automatic differentiation. The PDE residual is enforced at randomly sampled collocation points. This approach is mesh-free and demonstrated on benchmark problems including Burgers’ equation, the nonlinear Schrödinger equation, and Navier–Stokes equations.

Discrete-Time PINNs — To address scalability limitations, implicit Runge–Kutta time-stepping schemes are embedded within the neural network structure. This allows larger time steps while maintaining numerical stability and accuracy, creating a bridge between classical numerical methods and deep learning.

Additionally, the framework extends to inverse problems, where unknown physical parameters are learned directly from data by treating them as trainable variables. The study demonstrates successful identification of PDE parameters and reconstruction of hidden quantities such as pressure fields in fluid flow problems.

References

1. M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

4 Physics-Informed Neural Network for the Nonlinear Schrodinger Equation

4.1 Abstract

The Nonlinear Schrodinger equation (NLSE) is a partial differential equation which is involved in different fields of physics such as quantum mechanics, nonlinear optics, plasma physics, and Bose–Einstein condensates. It depicts the evolution of the nonlinear wave packets and supports a shape of soliton in its solutions.

In this work, a Physics-Informed Neural Network (PINN) is created to solve the one-dimensional NLSE in a mesh-free manner. Instead of discretizing space and time using traditional numerical methods, PINN enforce the governing physics rules dierectly into the nueral network traing process. The model learns about the space and the effect of time in it from the intial condition, boundary conditions, and by minimising the residual of the PDE. The final results predicted the soliton-type shape of NLSE by preserving its localized structure across time.

4.2 Introduction

The NLSE plays an important role in modeling wave propagation in nonlinear dispersive media. Unlike linear wave equations, the NLSE contains an aditional term which lead to the nonlinearisation of the wave which leads to complex phenomena such as soliton formation, wave localization, and stability over a long time.

The tradidional numerical method requires a fine meshes and small time steps to maintain stability and reproduce the expected results. A perfect descritisation can give the exact results, but such descritisations must be extremely small which increases the computational cost high. Also such methods follow a compounding error in each time step which will affect the output in a complex nonlinear problems.

PINNs offer an alternative approach. Instead of approximating the solution through grid-based discretization, a neural network is trained to predict the unknown solution function. The governing PDE is incorporated into the loss function using automatic differentiation. As a result, the network learns the physical law and the involved constraints.

This study aims to demonstrate that a PINN can accurately learn the full spatio-temporal solution of the one-dimensional NLSE using limited initial and boundary information.

4.3 Governing Equation

The one-dimensional nonlinear Schrodinger equation considered in this work is given by

$$ih_t + \frac{1}{2}h_{xx} + |h|^2h = 0, \quad (2)$$

defined over the spatial and temporal domains $x \in [-5, 5]$, $t \in [0, \pi/2]$. The initial condition is $h(0, x) = 2 \operatorname{sech}(x)$, and periodic boundary conditions are imposed such that $h(t, -5) = h(t, 5)$, $h_x(t, -5) = h_x(t, 5)$.

Since the solution $h(t, x)$ is complex-valued, it is decomposed into real and imaginary components:

$$h(t, x) = u(t, x) + iv(t, x), \quad (3)$$

where $u(t, x)$ and $v(t, x)$ represent the real and imaginary parts, respectively. The neural network is designed to learn both components simultaneously.

4.4 Methodology

Neural Network Architecture The PINN takes spatial and temporal coordinates (x, t) as inputs and outputs the corresponding real and imaginary parts of the solution. The architecture consists of five hidden layers with one hundred neurons in each layer. The hyperbolic tangent (tanh) activation function is used throughout the hidden layers, as it provides smooth approximations suitable for representing continuous physical fields.

The network parameters are optimized such that the predicted function satisfies the initial condition, the periodic boundary conditions, and the governing differential equation.

Physics-Informed Loss Function The total loss function combines two components:

- Initial condition loss, ensuring agreement with the prescribed initial wave profile.
- PDE residual loss, minimizing the residual of the nonlinear Schrodinger equation at randomly sampled collocation points in the space-time domain.

Automatic differentiation is employed to compute the required derivatives h_t , h_x , and h_{xx} . This enables the exact evaluation of derivatives with respect to network inputs, avoiding numerical approximation errors.

The network is trained using the Adam optimizer. During training, collocation points are sampled throughout the domain to ensure that the learned function satisfies the governing equation globally.

4.5 Results

The trained PINN successfully reconstructs the full spatio-temporal evolution of the nonlinear wave. The magnitude of the complex solution is defined as

$$|h(t, x)| = \sqrt{u^2 + v^2}. \quad (4)$$

Spatio-Temporal Evolution

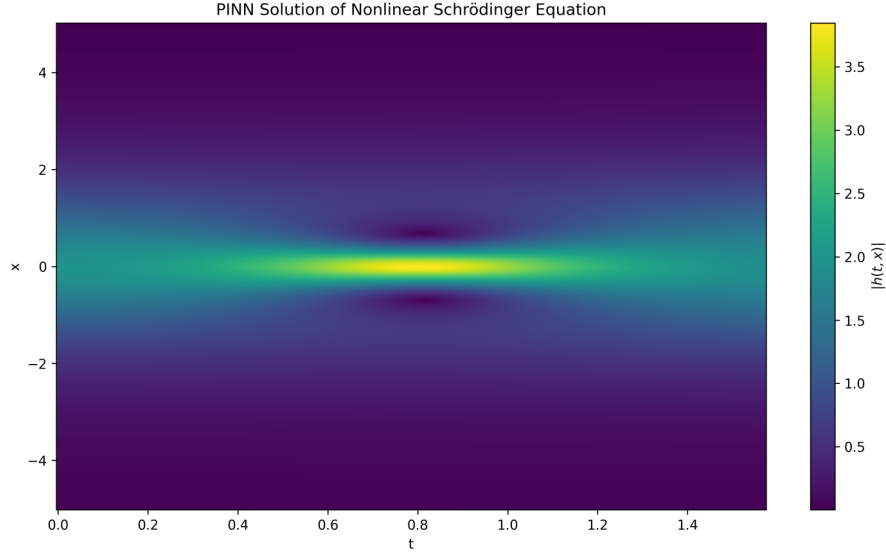


Figure 1: Spatio-temporal evolution of the magnitude $|h(t, x)|$ predicted by the PINN. The solution shows a stable structure centered at $x=0$ over the entire time interval.

The figure clearly depicts the stable nature of NLSE. The wave has a maximum at the centre and it gradually fades when moved away from the center. Furthermore, the structure of the wave is well maintained. The gradual fading of the wave when the time changes without any oscillations or irregular behaviours shows that the implemented PINN successfully reproduced the result.

Time-Slice at $t = 0.8$

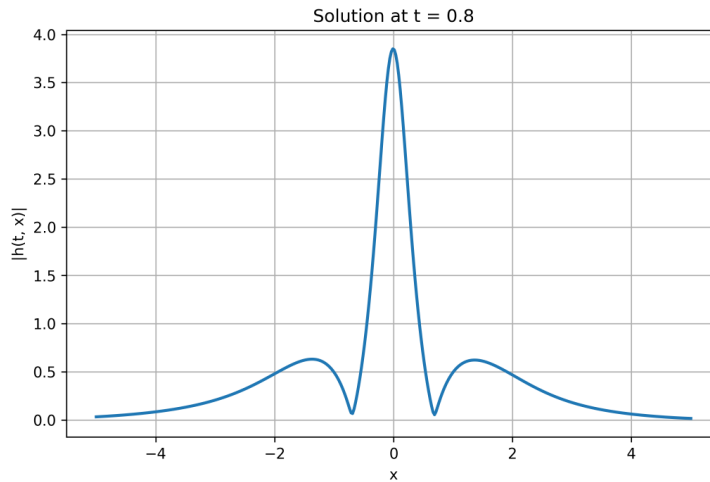


Figure 2: Graphical representation of the predicted magnitude $|h(t, x)|$ at time $t = 0.8$. The solution maintains a sharp central peak with symmetric decay on both sides.

The graphical representation at $t=0.8$ highlights the localised nature of the solution. Overall, the results confirm that the PINN captures both the nonlinear interaction and dispersive behavior inherent to the Schrodinger equation. A clear central peak is visible near $x = 0$, accompanied by smaller side lobes with a symmetrical decay towards the boundaries. Here also the graph is smooth without any oscillations or disparities, confirming that the trained PINN generalizes well across time and inherited the NLSE traits perfectly. Overall, the results confirm that the PINN captures both the nonlinear interaction and dispersive behavior inherent to the NLSE.

4.6 Discussion

The results indicate that PINNs can effectively predict complex-valued nonlinear PDEs without relying on mesh-based discretization. The solution remains stable throughout the entire time interval and respects periodic boundary conditions.

This approach is better when the available data is limited and the prediction becomes impossible with normal neural network. The normal neural network predicts the results by recognizing the patterns from data. But when data becomes scarce, PINNs which combine the pattern recognition skill of neural network with the learning enforced through physical laws come into the picture. The work discussed before ensures the viability of PINNs in complex problems.

However, training such models can be computationally demanding. The convergence behavior depends on hyperparameter selection, collocation sampling, network architecture, and network initialization. Scaling this approach to higher spatial dimensions or more complex systems requires additional optimization strategies which is discussed in further sections of this Dissertation.

4.7 Conclusion

This study demonstrates that a PINN can successfully solve the one-dimensional nonlinear Schrodinger equation in a mesh-free manner. By incorporating the governing PDE equation into the loss function and leveraging automatic differentiation, the network perfectly recreated the full spatio-temporal evolution of a nonlinear wave from the limited initial and boundary information.

This result validates PINNs as a powerful alternative framework for scientific computing. They provide a flexible and Physically consistent approach for solving nonlinear PDEs, particularly the cases where traditional discretization methods become computationally intensive or difficult to implement.

5 Physics-Informed Neural Network for the Diffusion Equation and Comparison with Explicit Finite Difference Method

5.1 Introduction

The diffusion equation is one of the most fundamental partial differential equations in physics and engineering. It models heat transfer, mass diffusion, particle transport, and various parabolic processes.

The one-dimensional diffusion equation is given by:

$$u_t = Du_{xx} \quad (5)$$

where, $u(x, t)$ is the scalar field (temperature or concentration), and D is the diffusion coefficient.

In this chapter, the diffusion equation is solved using:

1. Physics-Informed Neural Network (PINN),
2. Explicit Finite Difference Method (FDM).

The goal is not only to solve the equation, but to analyze and compare both approaches.

5.2 Physics-Informed Neural Network for the Diffusion Equation

5.2.1 Abstract

The diffusion equation is a fundamental partial differential equation that models processes such as heat transfer, mass transport, and concentration spreading in physical systems. Traditional numerical approaches rely on spatial discretization, which may introduce stability constraints and numerical errors.

In this work, a Physics-Informed Neural Network (PINN) is developed to solve the one-dimensional diffusion equation in a mesh-free manner. The governing physical law is embedded directly into the neural network loss function, allowing the model to learn the continuous solution across space and time. The results validate that PINN accurately predicted the smooth behavioral characteristics of diffusion without depending on explicit discretization.

5.2.2 Introduction

Diffusion processes describe how heat, particles, or concentration fields spread over time in space. The classical diffusion equation provides a mathematical formulation of this phenomenon which plays an important role in engineering, physics, and applied mathematics.

Conventional numerical methods such as finite difference method require discretizing the domain into grids. These method introduces several errors due to this discretizing nature like cumulative truncation error. On the other hand, PINNs provide an alternative framework. Instead of approximating the solution through discretisation, a neural network is trained to

represent the continuous solution function. The governing differential equation is enforced through the loss function using automatic differentiation. As a result, the network learns a solution that satisfies both the physics and the involved initial and boundary conditions.

This study shows that a PINN can successfully learn the full space-time solution of the diffusion equation while maintaining the physical consistency.

5.2.3 Governing Equation

The one-dimensional diffusion equation is given by

$$u_t = Du_{xx} \quad (6)$$

where $u(x, t)$ is the concentration field and D is the diffusion coefficient.

A Gaussian profile is introduced as the initial condition, and the boundary conditions are homogeneous Neumann boundary condition, in other words the boundaries are completely reflective in nature. $u_x = 0$, which represents zero flux across the boundaries.

5.2.4 Methodology

Neural Network architecture The PINN takes two inputs which consists of the spacial coordinate (x) and the time coordinate (t) and will predict concentration $u(x, t)$ as output.

The architecture consists of 4 hidden layers with 64 neurons per hidden layer. tanh is the activation functions are used with a fully connected feedforward structure.

The tanh activation function is selected because it produces smooth outputs and is better for approximating continuous physical fields.

Physics-Based Loss function The total loss function is defined as

$$L = \lambda_{\text{PDE}} L_{\text{PDE}} + \lambda_{\text{IC}} L_{\text{IC}} + \lambda_{\text{BC}} L_{\text{BC}}. \quad (7)$$

PDE Residual Loss The governing equation is rewritten in residual form as

$$f = u_t - Du_{xx}. \quad (8)$$

The network reduces the mean square value of the residual by training at randomly sampled collocation points which ensures the satisfaction of diffusion equation across the domain.

Initial Condition Loss The network prediction at $t = 0$ must match the initial condition introduced which is Gaussian here.

Boundary Condition Loss The homogeneous Neumann condition $u_x = 0$ must be enforced at the spatial boundaries.

5.2.5 Training Strategy

The training is done using Adam optimiser which intern used 10,000 epochs for better prediction. Random collocation points are used for training across whole the domain. During early time, more weightage is given to the initial condition loss function to maintain the intial Gaussian profile. Since the nueral netwrok produce a continuous function as its output, we will get a continuous function which is differentiable across the whole domain.

5.2.6 Results

Space-Time Evolution

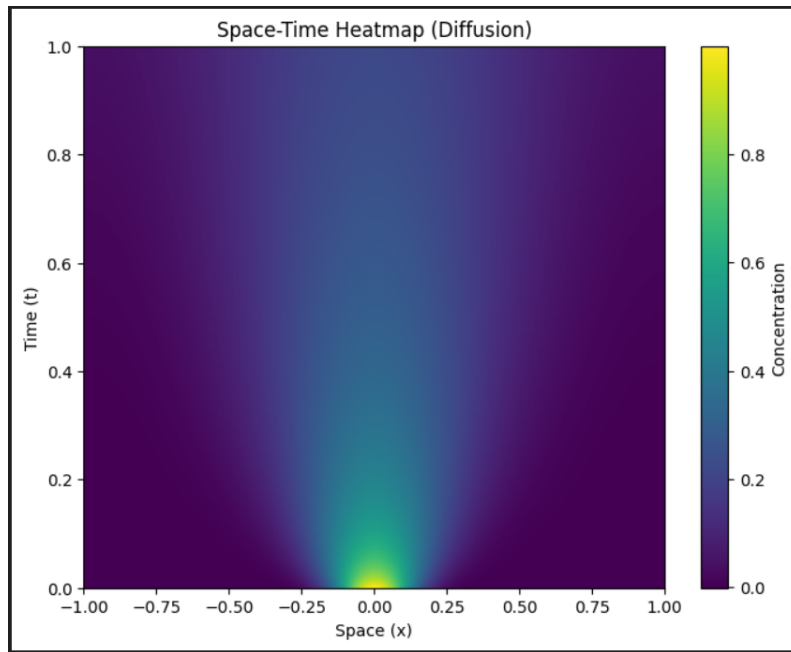


Figure 3: Space-time heatmap of the concentration field $u(x,t)$. The prediction clearly depicts the gradual decrease of its peak amplitude, illustrating the diffusion process.

The space-time heat map shows the gradual decrease of the peak amplitude and spreading out of the Gaussian profile which is the inherit character of a diffusion equation.

Diffusion at Multiple Time Instances

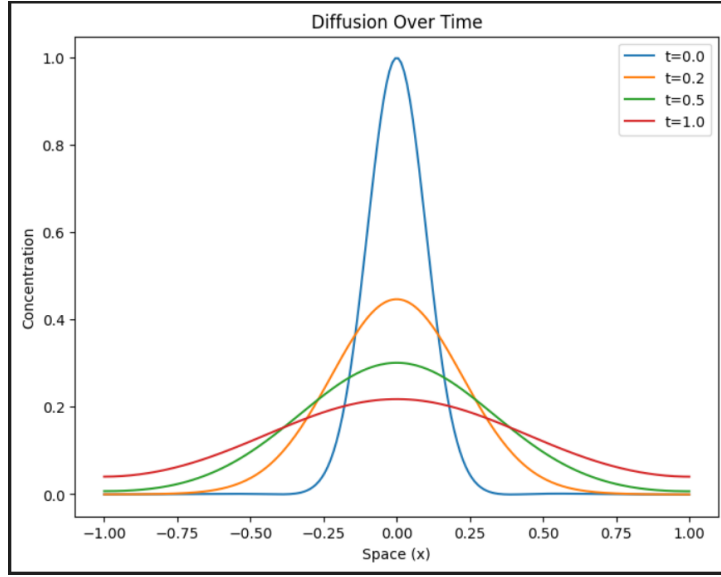


Figure 4: Spatial concentration profiles at multiple time instances ($t = 0.0, 0.2, 0.5, 1.0$). The Gaussian peak progressively broadens and decreases in amplitude over time.

The spatial profile gives a more visual understanding of the diffusion equation. The profile shows that the PINN correctly predicted the results by clearly representing the Gaussian curve spreading as time moves forward.

5.3 Explicit Finite Difference Method for the Diffusion Equation

5.3.1 Abstract

Finite difference method(FDM) is one of the most accepted and widely used explicit method used to solve PDEs among the classical numerical approaches. In this method we approximate derivatives using discrete differences in space and time.

In this section, the explicit FDM is implemented to solve the one-dimensional diffusion equation with a Gaussian initial condition and Neumann boundary conditions. The numerical behavior, stability requirements, and computational characteristics of the method are discussed in detail.

5.3.2 Governing Equation

The one-dimensional diffusion equation is given by

$$u_t = Du_{xx}, \quad (9)$$

where $u(x, t)$ represents concentration and D is the diffusion coefficient.

The domain is discretized into spatial grid points and time levels. The explicit finite difference scheme uses a forward difference approximation in time and a central difference approximation in space.

5.3.3 Numerical Scheme

The time derivative is approximated using a forward difference:

$$u_t \approx \frac{u_i^{n+1} - u_i^n}{\Delta t}, \quad (10)$$

and the second spatial derivative is approximated using a central difference:

$$u_{xx} \approx \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}. \quad (11)$$

Substituting these approximations into the diffusion equation yields:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = D \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}. \quad (12)$$

Rearranging, we obtain the explicit update formula:

$$u_i^{n+1} = u_i^n + r (u_{i+1}^n - 2u_i^n + u_{i-1}^n), \quad (13)$$

where $r = \frac{D\Delta t}{\Delta x^2}$.

This formula allows the solution at the next time step to be computed explicitly from known values at the current time level.

5.3.4 Stability Condition

The explicit scheme is conditionally stable. Stability requires $r \leq \frac{1}{2}$. This restriction is commonly referred to as the Courant–Friedrichs–Lewy (CFL) condition. If this condition is violated, numerical errors grow rapidly and the solution becomes unstable.

As a result, the time step Δt must be chosen sufficiently small relative to the spatial grid size Δx . For fine spatial grids, this leads to very small time steps and increases computational cost.

5.3.5 Implementation Details

The FDM method is implemented by considering the same constraints we used in the PINN. The initial condition consists of the Gaussian profile, the boundary conditions are reflective with nuemann boundaries. The method involves discretizing the space and time which leads to a time-stepping discretization. The Neumann boundary condition is enforced by approximating the zero-gradient constraint at the domain boundaries.

5.3.6 Results

Space-Time Evolution

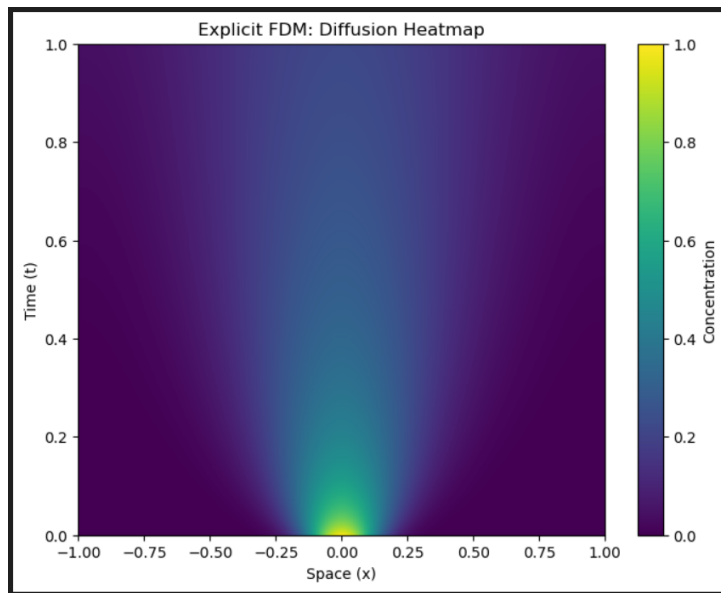


Figure 5: Space-time heatmap obtained using the explicit FDM. The Gaussian profile gradually spreads and its amplitude decreases over time, consistent with diffusion behavior.

The space-time heat map shows the gradual decrease of the peak amplitude and spreading out of the Gaussian profile which is the inherent character of a diffusion equation.

Time-Slice Profiles

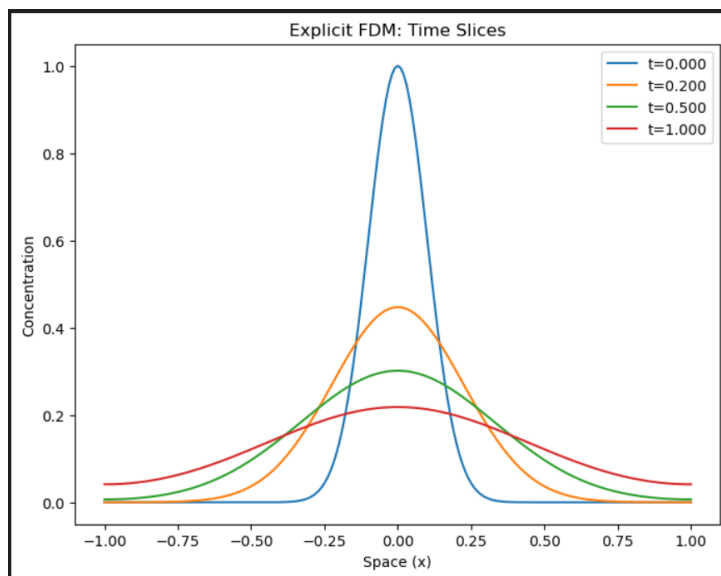


Figure 6: Spatial concentration profiles at multiple time instances using the explicit finite difference method. The Gaussian peak progressively widens and decreases in amplitude.

The spatial profile gives a more visual understanding of the diffusion equation. The profile shows that the PINN correctly predicted the results by clearly representing the Gaussian curve spreading as time moves forward

The results confirm that the explicit method provides accurate approximations when the stability condition is satisfied.

5.4 Comparison: Physics-Informed Neural Network (PINN) vs Explicit Finite Difference Method (FDM)

5.4.1 Quantitative Comparison Between PINN and Explicit FDM

The following figure represents the comparison of PINN and FDM solution at different time instances ($t = 0.00, 0.20, 0.50, 1.00$).

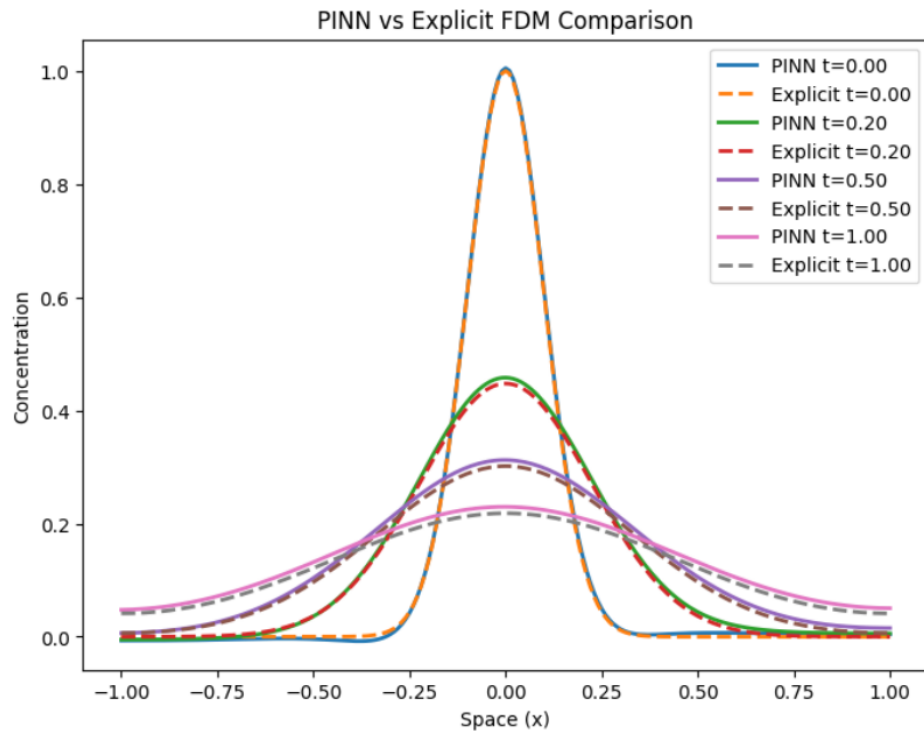


Figure 7: Comparison between PINN (solid lines) and Explicit FDM (dashed lines) solutions at multiple time instances. The curves show strong agreement across all time snapshots.

When $t = 0$ both methods coincide with each other. As time moves forward, a gradual spread happens making the Gaussian profile more flat with decreased amplitude. This is the characteristic behavior of diffusion. Hence Both PINN and Explicit FDM produced almost identical behaviour and reproduced the expected result. This shows that both PINN and FDM are capable of handling such problems.

5.4.2 Observation

Explicit FDM is a mesh based discretisation method while PINN is a mesh free approach. The explicit FDM is conditionally stable. It requires the parameter $r = \frac{D\Delta t}{\Delta x^2}$ to satisfy $r \leq 0.5$. If this Courant–Friedrichs–Lewy (CFL) condition is violated, numerical instability occurs and the solution diverges.

PINNs do not have an explicit stability restriction related to time-step size. Since there is no time marching, the concept of CFL stability does not apply. Instead, stability depends on optimization convergence. Poor training, insufficient collocation points, or improper loss balancing may lead to inaccurate solutions, but not classical numerical instability.

The explicit FDM is computationally inexpensive per time step. However, because it advances step by step in time, the total computational cost grows linearly with the number of time steps and spatial grid resolution.

PINNs require a computationally expensive training phase due to iterative optimization over many epochs. Training can be significantly slower than running a classical solver. However, once trained, evaluating the solution at new points is extremely fast. This makes PINNs attractive for applications requiring repeated queries.

5.4.3 Continuation

Now we need to investigate more about the utilisation of PINN in complex physics problems as well. The following chapters of the project will be a complete comparative analysis of PINNs with FDM in a diverse physical systems for a better understanding and conclusion.

6 Transport Dynamics and Boundary Variations in 1D Diffusion

6.1 Abstract

While diffusion often involves unrestricted spread of concentration field, real life diffusion problems will not be this symmetric. Real world physical systems often involve asymmetric boundary conditions which strongly influence the transport dynamics. In this section, we will continue our comparison of PINN with FDM in the 1-D diffusion problems with varying boundary conditions. Two different systems are investigated for this purpose:

1. A domain with one absorbing boundary and one reflective boundary.
2. A domain with two absorbing boundaries.

To characterize the transport behavior, physical observables such as the Survival Probability and Mean Squared Displacement are computed throughout the section.

The results demonstrate that both PINNs and explicit FDM accurately reproduced the expected normal diffusion dynamics.

6.2 Introduction

In statistical mechanics there is a very high importance to the domain boundaries of the system. Boundaries may absorb, reflect, or partially absorb particles which can influence the long-term transport behavior of the system

Two physical quantities which are used to classify transport dynamics are:

1. **Survival Probability**, which measures the fraction of particles remaining inside the domain.
2. **Mean Squared Displacement (MSD)**, which characterizes the spreading of the concentration field.

The behavior of these quantities will be evaluated for the classification of the transport dynamics into:

1. Subdiffusion
2. Normal diffusion
3. Superdiffusion or ballistic

PINN enforce the boundary conditions by directly optimising the loss function. FDM on the other hand requires different kind of approach based on the type of boundary. When the boundary is reflective, it needs ghost node to enforce zero flux conditions.

In this section, both PINNs and explicit FDM are implemented over the domain to study diffusion under different boundary configurations. The resulting dynamics are analyzed through late-time power-law scaling of the MSD and Survival Probability.

6.3 Mathematical Formulation and Metrics

The one-dimensional diffusion equation considered in this study is

$$u_t = Du_{xx}, \quad (14)$$

where $u(x, t)$ represents the concentration field and D is the diffusion coefficient.

The equation is solved over the asymmetric spatial domain $x \in [-5, 25]$, and the temporal domain $t \in [0, 900]$. The diffusion coefficient is chosen as $D = 0.1$.

The initial condition is prescribed as a Gaussian pulse centered at $x_0 = 0.0$ with width $\sigma = 1.0$:

$$u(x, 0) = \frac{1}{\sigma\sqrt{\pi}} \exp\left(-\frac{(x-x_0)^2}{\sigma^2}\right). \quad (15)$$

6.3.1 Transport Metrics

To analyze the transport dynamics quantitatively, two physical observables are computed throughout the simulation.

Survival Probability The Survival Probability $S(t)$ measures the total concentration remaining inside the domain at time t and is defined as

$$S(t) = \int u(x, t) dx. \quad (16)$$

For absorbing boundaries, particles gradually leave the domain, causing $S(t)$ to decrease over time.

Mean Squared Displacement The Mean Squared Displacement (MSD) characterizes the spatial spreading of the surviving concentration profile and is defined as

$$MSD(t) = \int (x - x_0)^2 \frac{u(x, t)}{S(t)} dx. \quad (17)$$

The MSD provides a measure of how rapidly the diffusion front expands with time.

6.3.2 Transport Classification

The transport regime is identified by fitting power-law relationships to the late-time behavior of the computed observables:

$$MSD(t) \propto t^\alpha, \quad (18)$$

and

$$S(t) \propto t^{-\beta}. \quad (19)$$

The exponents α and β are extracted using linear fitting in log-log space.

For normal diffusion, the theoretical expectations are $\alpha \approx 1.0$ and $\beta \approx 0.5$.

Deviation from these values would indicate different transport behavior such as subdiffusion or superdiffusion.

6.4 One-Dimensional Asymmetric Bounded Diffusion: PINN vs. Explicit FDM

6.4.1 Abstract

In this section we investigate one-dimensional diffusion within an asymmetric bounded domain having one perfectly absorbing (Dirichlet) boundary and one perfectly reflective (Neumann) boundary.

Such mixed boundary conditions generate nontrivial transport dynamics involving simultaneous mass depletion and zero-flux reflection, providing an ideal framework for comparing classical numerical solvers against modern physics-informed deep learning approaches.

Both an Explicit Finite Difference Method (FDM) and a mesh-free Physics-Informed Neural

Network (PINN) are created and the transport regime is quantitatively classified using the Survival Probability: $S(t)$. For symmetric diffusion systems, we would have used the Mean Squared Displacement (MSD) for the classification of transport. However in the single absorbing boundary systems the long-time statistical behavior of survival probability is following a power-law decay:

$$S(t) \propto t^{-\beta}, \quad (20)$$

with a theoretical normal-diffusion exponent $\beta \approx 0.5$.

The comparative analysis demonstrates that both frameworks successfully captured the asymmetric dynamics and classified the system as normal diffusion.

The extracted decay exponents are $\beta_{\text{PINN}} = 0.457$, and $\beta_{\text{FDM}} = 0.432$.

The results highlights the differences between mesh based numerical method and continuous neural network, particularly in the treatment of mixed boundary conditions, error propagation, and long time transport analysis.

6.4.2 Governing Equation and Physical Setup

The 1-D diffusion equation which the system follows:

$$u_t = Du_{xx}, \quad (21)$$

where $D = 0.1$ is the diffusion coefficient. The spatial domain is asymmetrically defined as $x \in [-5, 25]$, and the temporal evolution is simulated over the extended interval $t \in [0, 900]$. The initial condition is a normalized Gaussian pulse centered at $x_0 = 0.0$ with width parameter $\sigma = 1.0$:

$$u(x, 0) = \frac{1}{\sigma\sqrt{\pi}} \exp\left(-\frac{(x - 0.0)^2}{\sigma^2}\right). \quad (22)$$

The system contains asymmetric mixed boundary conditions with a absorbing Boundary at Left, $u(t, -5) = 0$ which removes all concentration reaching the boundary and a reflective Boundary right $u_x(t, 25) = 0$, which enforces zero mass flux across the boundary.

6.4.3 Numerical Benchmark: Explicit FDM

Numerical Scheme The classical numerical benchmark utilizes a Forward-Time Central-Space (FTCS) explicit finite difference scheme.

The continuous spatial and temporal domains are discretized using $\Delta x = 0.1$, and $\Delta t = 0.04$. To ensure numerical stability, the system should satisfy the CFL stability condition:

$$\alpha = \frac{D\Delta t}{\Delta x^2} \leq 0.5. \quad (23)$$

Substituting the selected discretization parameters gives the stable parameter $\alpha = 0.40$.

Interior grid points evolve according to the standard FTCS update rule:

$$u_i^{n+1} = u_i^n + \alpha (u_{i+1}^n - 2u_i^n + u_{i-1}^n). \quad (24)$$

Boundary Condition Implementation

Absorbing Boundary The Dirichlet condition at $x = -5$ is implemented directly by clamping the first grid node to zero at every time step: $u_0^{n+1} = 0$.

Reflective Boundary The Neumann reflective boundary requires $u_x = 0$. This condition is implemented using a ghost node outside the physical computational domain.

If the final physical node corresponds to index N , the ghost node is defined through symmetry:

$$u_{N+1}^n = u_{N-1}^n. \quad (25)$$

Substituting this relation into the FTCS equation yields the reflective-boundary update equation:

$$u_N^{n+1} = u_N^n + 2\alpha (u_{N-1}^n - u_N^n). \quad (26)$$

6.4.4 PINN Architecture and Training Strategy

Network Architecture To solve the diffusion equation without relying on discrete spatial mesh, the PINN represents the concentration field $u(x, t)$ as a continuous differentiable neural function.

The architecture contains 3 hidden layers with 64 neurons per hidden layer with $\tanh$ activation functions.

The smoothness of the $\tanh$ activation will help in accurate evaluation of continuous derivatives u_t , u_x , u_{xx} through automatic differentiation.

Loss Function Formulation The total physics-informed loss function is defined as:

$$L_{\text{Total}} = L_{\text{PDE}} + \lambda_{\text{BC}} (L_{\text{BC, left}} + L_{\text{BC, right}}) + \lambda_{\text{IC}} L_{\text{IC}}. \quad (27)$$

The individual loss terms are:

PDE Residual Loss

$$L_{\text{PDE}} = \text{MSE}(u_t - Du_{xx}, 0). \quad (28)$$

Absorbing Boundary Loss

$$L_{\text{BC, left}} = \text{MSE}(u(-5, t), 0). \quad (29)$$

Reflective Boundary Loss

$$L_{\text{BC, right}} = \text{MSE}(u_x(25, t), 0). \quad (30)$$

Initial Condition Loss

$$L_{\text{IC}} = \text{MSE}(u(x, 0), u_{\text{initial}}). \quad (31)$$

Strong penalty weights $\lambda_{\text{BC}}, \lambda_{\text{IC}} \gg 1$ ensure strict enforcement of the physical constraints.

Collocation Point Sampling Strategy To preserve the sharp gradients of the initial Gaussian pulse, a targeted collocation-point sampling strategy is used.

The initial-condition points are divided into two groups:

1. Half sampled uniformly throughout the domain.
2. Half sampled from a normal distribution concentrated near $x = 0$.

This prevents excessive smoothing of the peak during optimization.

Multi-Phase Optimization Strategy Training a PINN to solve a PDE involves dealing a highly complex non-convex loss landscape where the data loss and physics loss directly compete.

To stabilize optimization and prevent gradient problems, a three-phase training process is executed.

Phase 0: Initial-Condition Pretraining The network is first trained using L_{IC} for 2000 epochs using the Adam optimizer.

This phase forces the network to accurately memorize the sharp Gaussian pulse before the PDE residual is introduced.

Without this pretraining phase, the network may artificially flatten the initial peak to reduce the second-derivative penalty associated with u_{xx} .

Phase 1: Full Physics Training (Adam) Once the initial state is implemented, the complete composite loss L_{Total} is activated. The model is optimized using Adam optimiser for 10,000 epochs with learning rate of 10^{-3} . Adam functions are an adaptive first-order optimizer capable of balancing the competing gradients from the interior PDE physics, boundary constraints, and initial-condition enforcement.

Phase 2: L-BFGS Refinement Although Adam effectively locates the general minimum point in convex loss function, its stochastic nature limits high-precision convergence.

To extract accurate long-time physical scaling laws such as β , the residual error must be minimized to near machine precision.

The optimization therefore switches to the quasi-Newton L-BFGS algorithm, which approximates the Hessian matrix and fine tunes the network weights.

The optimizer is applied for up to 1000 iterations to maximize physical understanding.

6.4.5 Results and Comparative Analysis

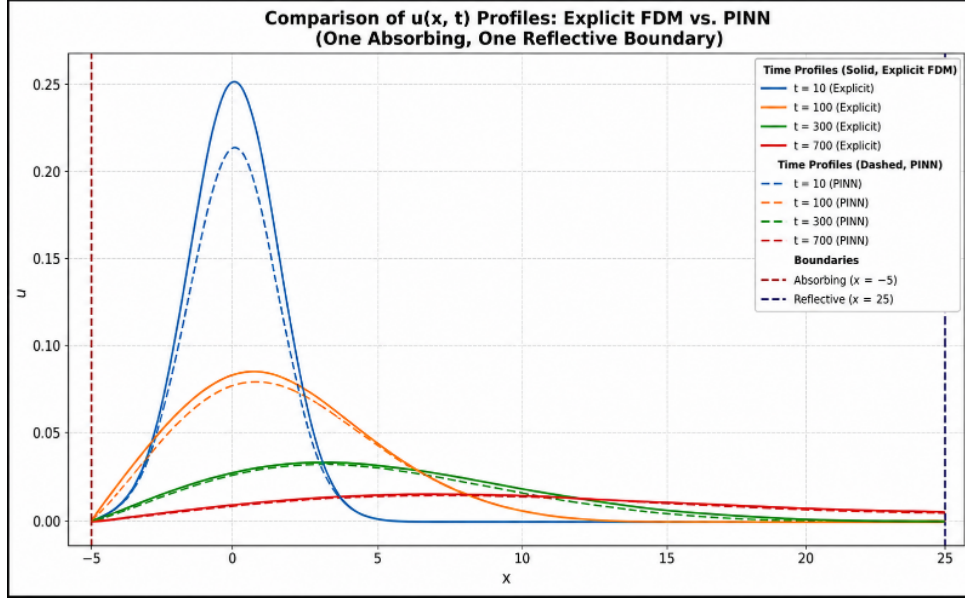


Figure 8: Comparison between PINN(dashed) and FDM(solid) at $t=10, 100, 300, 700$.

Spatio-Temporal Evolution Both models accurately reproduced the asymmetric transport behavior.

The left side of the Gaussian pulse continuously decays toward zero as particles are absorbed at $x = -5$. Near the reflective boundary at $x = 25$, the concentration profile becomes more flat.

While the Explicit FDM relies on ghost-node approximations to enforce $u_x = 0$, the smoothness of the PINN representation allows the reflective behavior to emerge continuous across the domain.

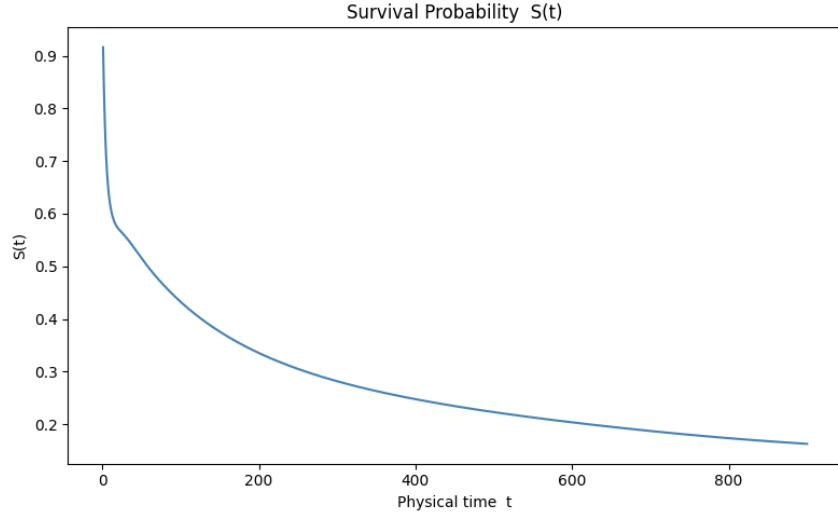


Figure 9: PINN Survival Probability $S(t)$ over physical time.

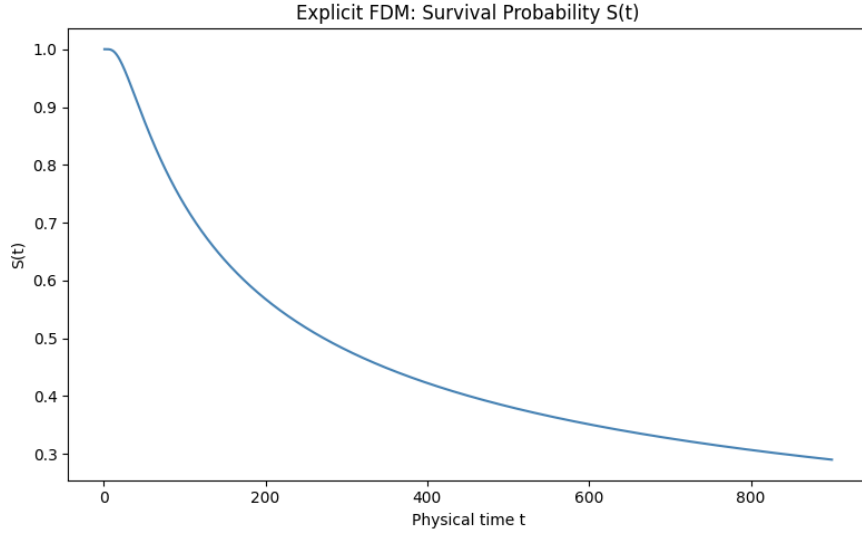


Figure 10: Explicit FDM Survival Probability $S(t)$ over physical time.

Survival Probability Analysis Both frameworks produce matching large-scale statistical behavior.

Initially, the survival probability decreases rapidly because the left-moving portion of the Gaussian pulse quickly reaches the nearby absorbing boundary.

At later times, the reflective boundary slows the decay by retaining particles inside the domain, but it must be noted that PINN failed to some extent to depict the initial decrease of survival probability. It shows a very sudden decrease of survival probability at initial time which deviates from the actual behaviour. Although PINN showed some disparity at the starting, the later regions are graphically represented in a more accurate manner compared to FDM.

Transport Classification via Log-Log Survival Probability Scaling Theoretical models for normal diffusion with a single absorbing boundary predict the scaling law:

$$S(t) \propto t^{-\beta}, \quad (32)$$

with $\beta \approx 0.5$.

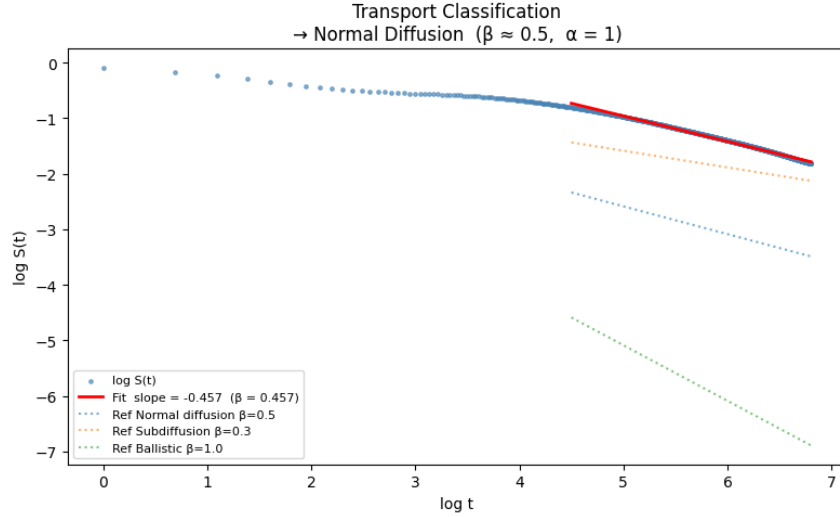


Figure 11: PINN transport classification via log-log survival probability analysis yielding $\beta = 0.457$.

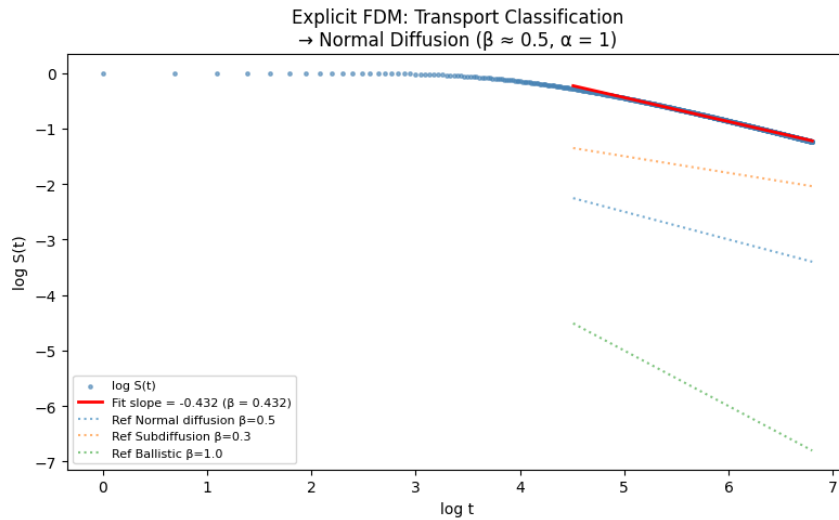


Figure 12: Explicit FDM transport classification via log-log survival probability analysis yielding $\beta = 0.432$.

Linear regression performed on the stable late-time regime yields $\beta_{\text{PINN}} = 0.457$, and $\beta_{\text{FDM}} = 0.432$.

Both frameworks therefore correctly classified the system as normal diffusion.

The PINN exponent remains marginally closer to the theoretical baseline $\beta \approx 0.5$.

6.4.6 Discussion and Conclusion

The comparative analysis highlights major differences between mesh-based numerical method and continuous neural network.

Boundary Condition Enforcement The Explicit FDM relies on numerical constructions such as ghost nodes to enforce Neumann derivatives.

Meanwhile, PINN handles reflective boundaries through automatic differentiation, preserving the continuous mathematical structure of the governing boundary conditions.

Error Propagation and Accuracy The FDM error in the decay exponent which is, $\beta = 0.432$ arises from the accumulation truncation errors, ghost-node approximations, and sequential time stepping errors created over 20,000+ iterations.

However PINN does not have this kind of errors because of its optimisation techniques, which includes Adam + L-BFGS. Due to this, it avoids cumulative time stepping errors and preserves long time transport scaling laws $\beta = 0.457$.

However the difference in β value proved that PINN must be trained more with extremely large number of collocation points with more complex network architecture, which further raises the issue of computational cost.

Computational Cost and Stability FDM is constrained with CFL stability limit. Although each update is computationally inexpensive, the method remains tied to this stability restriction.

The PINN completely free from the CFL constraints.

However, this flexibility requires computational training with multi-phase optimization, large collocation datasets, long neural-network training cycles, and navigation through a highly non-convex optimization landscape. Which account for high computational cost compared to the FDM method.

6.5 Case 2 :One-Dimensional Diffusion with Two Absorbing Boundaries: PINN vs. Explicit FDM

6.5.1 Abstract

IN this section we investigate 1-D diffusion within a bounded domain containing two perfectly absorbing boundaries.

Here unlike the previous case, when both boundaries are dirichlet, the mass reduction happens continuously which results in long time exponential survival decay. By using MSD this physical system is still an ideal benchmark for the continuation of our comparison.

The MSD is used to classify the transport dynamics through power law scaling:

$$MSD(t) \propto t^\alpha, \quad (33)$$

where normal diffusion predicts $\alpha \approx 1$. In addition, the late time survival probability exhibits exponential decay.

The comparative analysis demonstrates that both frameworks successfully reproduce the expected statistical mechanics of bounded diffusion.

The MSD exponents are $\alpha_{\text{PINN}} = 0.943$, and $\alpha_{\text{FDM}} = 0.963$. Both approaches also reproduced the theoretically predicted late-time exponential decay of the survival probability.

6.5.2 Governing Equation and Physical Setup

The physical system is governed by the 1-D diffusion equation $u_t = Du_{xx}$, where the diffusion coefficient is $D = 0.1$. The spatial domain is asymmetrically defined as $x \in [-5, 25]$, and the time interval of consideration is $t \in [0, 900]$.

The initial condition is a normalized Gaussian pulse centered at

$$x_0 = 0.0$$

with width parameter $\sigma = 1.0$ with its concentration profile given by:

$$u(x, 0) = \frac{1}{\sigma\sqrt{\pi}} \exp\left(-\frac{(x - 0.0)^2}{\sigma^2}\right). \quad (34)$$

The domain has two perfectly absorbing walls corresponding to homogeneous Dirichlet conditions: $u(t, -5) = 0$, $u(t, 25) = 0$. Any concentration reaching the boundary is permanently removed from the system.

6.5.3 Numerical Benchmark: Explicit FDM

Numerical Scheme The numerical method is constructed using a Forward-Time Central-Space (FTCS) explicit finite difference scheme.

The spatial domain is discretized with $\Delta x = 0.1$. To ensure numerical stability, the time step must satisfy the Courant–Friedrichs–Lewy (CFL) condition:

$$\alpha = \frac{D\Delta t}{\Delta x^2} \leq 0.5. \quad (35)$$

A selected time step is $\Delta t = 0.045$, producing the grid parameter $\alpha = 0.45$. The equation for FTCS update rule is:

$$u_i^{n+1} = u_i^n + \alpha (u_{i+1}^n - 2u_i^n + u_{i-1}^n). \quad (36)$$

Boundary Condition Enforcement Because both boundaries are homogeneous Dirichlet conditions, the implementation are straightforward with no ghost nodes involved.

At every time step, the boundary nodes concentration field must be zero. That is $u_0^{n+1} = 0$, $u_N^{n+1} = 0$.

Transport Observable Extraction To classify the transport dynamics, numerical integration across the spatial grid is performed.

The Survival Probability is equated into:

$$S(t) = \int u(x, t) dx, \quad (37)$$

which measures the total mass remaining inside the system.

The Mean Squared Displacement is:

$$MSD(t) = \int (x - x_0)^2 \frac{u(x, t)}{S(t)} dx. \quad (38)$$

6.5.4 Detailed PINN Architecture and Training Strategy

Network Architecture The PINN approximates the concentration field $u(x, t)$ as a continuous differentiable neural function. The network maps the spatio-temporal coordinates (x, t) to the scalar concentration output u

The architecture utilizes hidden layers with tanh activation functions. The smoothness of this activation function enables accurate evaluation of the derivatives u_t, u_x, u_{xx} through automatic differentiation.

Loss Function Formulation The network is trained by minimizing the composite Mean Squared Error (MSE) loss:

$$L_{\text{Total}} = L_{\text{PDE}} + \lambda_{\text{BC}} (L_{\text{BC, left}} + L_{\text{BC, right}}) + \lambda_{\text{IC}} L_{\text{IC}}. \quad (39)$$

PDE Residual Loss The governing equation residual is enforced throughout the interior domain:

$$L_{\text{PDE}} = \text{MSE}(u_t - Du_{xx}, 0). \quad (40)$$

Boundary Condition Loss The absorbing boundaries are enforced through:

$$L_{\text{BC, left}} = \text{MSE}(u(-5, t), 0), \quad (41)$$

and

$$L_{\text{BC, right}} = \text{MSE}(u(25, t), 0). \quad (42)$$

Initial Condition Loss The Gaussian pulse is imposed using:

$$L_{\text{IC}} = \text{MSE}(u(x, 0), u_{\text{initial}}). \quad (43)$$

Strong penalty weights $\lambda = 1000$ are applied, ensuring strict enforcement of both the boundary and initial conditions relative to the PDE residual.

Collocation Point Sampling Strategy To prevent excessive smoothing of the sharp Gaussian peak, collocation points are sampled just like in the previous one absorbing boundary PINN. This targeted sampling improves reconstruction of the initial profile.

Multi-Stage Optimization Strategy PINN Training involves navigating through a non-convex loss landscape where data loss competes with physics loss.

To ensure convergence, a multi phase training protocol is executed.

Phase 0: Initial Condition Pretraining The Adam optimizer is first applied for 2000 epochs while minimizing only L_{IC} . This forcibly embeds the Gaussian pulse into the neural representation before any physical temporal evolution is introduced, preventing the network from flattening the initial peak to artificially satisfy the spatial PDE derivatives.

Phase 1: Full Physics Training The complete composite loss function L_{Total} is activated. The network is trained for 10,000 epochs using the Adam optimizer with learning rate 10^{-3} .

Phase 2: L-BFGS Refinement Finally, the quasi-Newton L-BFGS optimizer is applied for 1000 iterations.

This second-order method fine-tunes the network weights, driving the residual loss toward near-zero tolerances and achieving high-precision convergence.

6.5.5 Results and Comparative Analysis

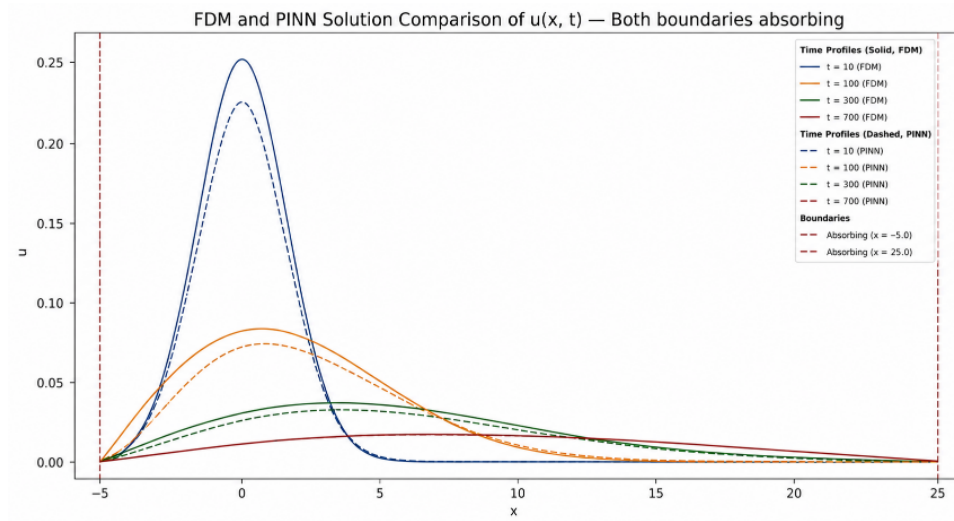


Figure 13: Comparison between PINN and FDM evaluated at $t = 10, 100, 300$, and 700 . The Gaussian pulse broadens over time while the concentration is continuously absorbed at both boundaries.

Spatio-Temporal Evolution (Snapshots) Both the PINN and the Explicit FDM successfully reproduce the bounded diffusion process.

As time goes by, the peak amplitude decreases while the distribution broadens spatially.

Because both boundaries act as perfect absorbing sinks, the concentration smoothly approaches zero at the edges in both models, resulting in continuous mass depletion from the physical system.

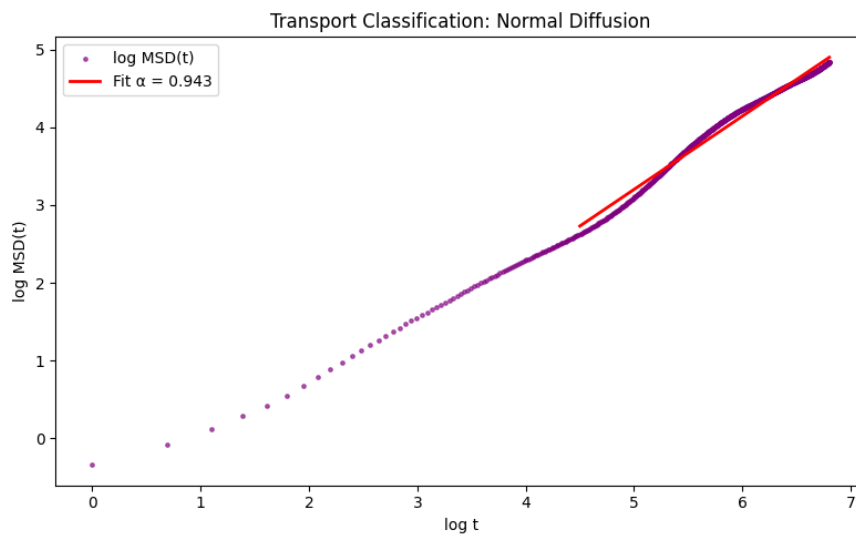


Figure 14: Transport classification via Mean Squared Displacement analysis for the PINN. The fitted scaling exponent of $\alpha = 0.943$ is consistent with normal diffusion.

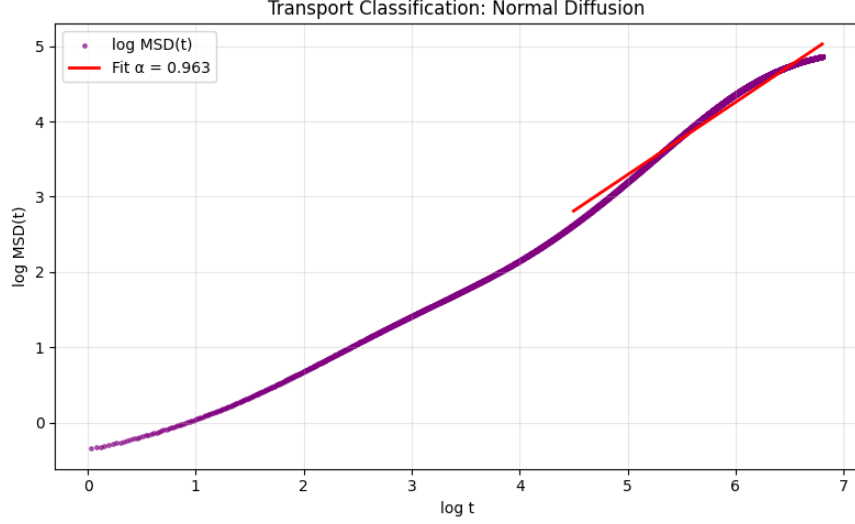


Figure 15: Transport classification via Mean Squared Displacement analysis for Explicit FDM. The fitted exponent $\alpha = 0.963$ confirms the normal diffusion regime.

Transport Classification via Mean Squared Displacement To classify the transport regime, the Mean Squared Displacement is analyzed in log-log space.

For normal diffusion:

$$MSD(t) \propto t^\alpha, \quad (44)$$

with $\alpha = 1$. The PINN regression analysis yields $\alpha_{\text{PINN}} = 0.943$ while the Explicit FDM regression analyse $\alpha_{\text{FDM}} = 0.963$. Both frameworks strongly confirm normal diffusion behavior.

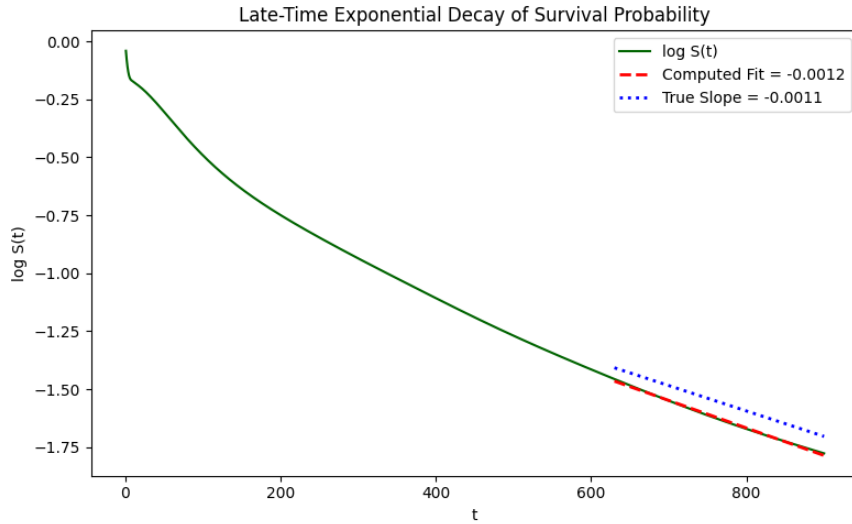


Figure 16: Late-time exponential decay of the PINN survival probability. The computed decay slope of -0.0012 closely matches the theoretical value of -0.0011 .

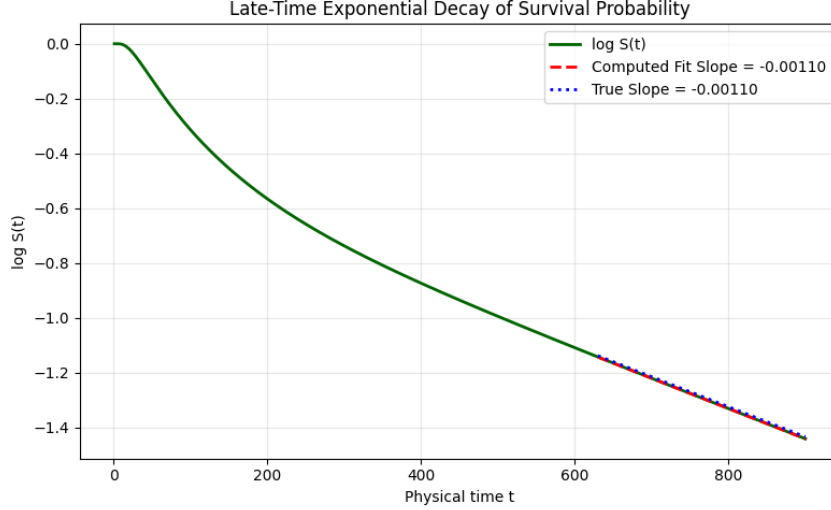


Figure 17: Late-time exponential decay of the survival probability for Explicit FDM. The computed decay slope matches the theoretical analytical prediction.

Late-Time Survival Probability Decay Diffusion confined by two absorbing boundaries produces exponential decay of the survival probability at late times.

The semi-logarithmic plot of $\log S(t)$ against linear time reveals a highly linear late-time regime. The PINN predicts a decay slope of -0.0012 , closely matching the theoretical analytical value -0.0011 . The Explicit FDM predicts a decay slope of -0.00110 , which agrees with the theoretical analytical value with near-perfect accuracy.

6.5.6 Discussion and Conclusion

Boundary Condition Enforcement Because both boundaries are homogeneous Dirichlet conditions, the FDM implementation is straightforward and does not require ghost nodes.

The PINN enforces the same constraints continuously through its boundary loss terms.

In both frameworks, the concentration smoothly approaches zero at the absorbing walls, confirming correct implementation of the physical constraints.

Error Propagation and Long-Time Accuracy We got FDM result of $\alpha_{\text{FDM}} = 0.963$. This small deviations come from the truncation error arising by the thousands of explicit time steps.

Conversely, the PINN produced $\alpha_{\text{PINN}} = 0.943$. Its deviation instead arise from incomplete optimization of the highly complex neural-network loss landscape.

Computational Cost and Stability FDM is constrained with CFL stability limit. Although each update is computationally inexpensive, the method remains tied to this stability restriction. The PINN completely free from the CFL constraints. However, this flexibility requires computational training with multi-phase optimization, large collocation datasets, long neural-network training cycles, and navigation through a highly non-convex optimization landscape. Which account for high computational cost compared to the FDM method.

Both the Explicit FDM and the PINN successfully reproduce the statistical mechanics of 1-D diffusion confined by two absorbing boundaries, accurately classifying the system as normal diffusion.

From the comparison its clear that the Explicit FDM remains the most computationally efficient and physically interpretable approach for low dimensional bounded diffusion systems.

6.6 Conclusion: PINN vs. Explicit FDM

This study demonstrates that both the FDM and the PINN successfully capture the transport dynamics of 1-D diffusion under mixed boundary conditions. However, the both approaches has distinct trade-offs.

- **FDM: Efficiency vs. Error Accumulation** THE FDM method is computationally lightweight and easy to implement. However, the error accumulation can be high in complex domain since errors like accumulated truncation errors are present in FDM due to its discretization approach. As a result, the extracted decay exponent $\beta = 0.432$ shows a small deviation from the theoretical expectation.
- **PINN: Accuracy vs. Computational Cost** PINN has a high computational cost due to its highly complex convex loss function optimisation. However, if an efficient PINN is trained with adequate amount of collocation, initial, and boundary points it will conserve the long term behaviour without much error

After careful observation Explicit FDM is found to outperform PINN in low dimensional domain since a highly trained PINN is an unwanted luxury which is pretty expensive to implement.

7 Two-Dimensional Bounded Diffusion: PINN vs. Explicit FDM

7.1 Abstract

In this section, we will move to a more complex physical system which consists of a 2-D domain with its boundaries perfectly absorbing. This problem has a great importance because it gives us a picture about how a model which depends on discretization perform when the domain evolve from 1-D to 2-D. Here also the comparison is done by the rigorous comparative analysis of MSD and Survival probability. The comparative results demonstrate that both frameworks successfully reproduce the results. Detailed analysis of the PINN architecture and weighted sampling strategy further reveals that FDM can perform well in higher dimensional domains also and PINN can closely reproduce the desired results along with FDM.

7.2 Governing Equation and Problem Setup

The 2-D diffusion equation governing the physical system is:

$$\frac{\partial u}{\partial t} = D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right). \quad (45)$$

The diffusion coefficient is $D = 0.1$. The spatial domain is a square defined by $x \in [-5.0, 5.0]$, and $y \in [-5.0, 5.0]$. The temporal evolution is simulated over $t \in [0, 50.0]$. The initial condition is a normalized 2-D Gaussian pulse centered at the origin, with width parameter $\sigma = 1.0$. The initial concentration field is therefore

$$u(x, y, 0) = \frac{1}{\pi\sigma^2} \exp \left(-\frac{(x-x_0)^2 + (y-y_0)^2}{\sigma^2} \right). \quad (46)$$

The domain is confined by four perfectly absorbing boundaries, enforcing Dirichlet conditions $u(\pm 5, y, t) = 0$, $u(x, \pm 5, t) = 0$.

7.3 Numerical Scheme (Explicit FDM)

The numerical method is designed by a 2-D Forward-Time Central- Space (FTCS) explicit finite difference method.

The spatial domain is discretized into a 200×200 grid, corresponding to approximate spatial resolutions $\Delta x \approx 0.0503$, $\Delta y \approx 0.0503$. To satisfy the 2-D CFL stability condition, the time step is restricted to $\Delta t \approx 0.00505$.

7.4 PINN Architecture and Training Strategy

To solve the 2-D diffusion equation, a specialized PINN architecture and rigorous optimization procedures were developed.

7.4.1 Network Architecture

The PINN utilizes a deep feedforward neural network with three inputs (x, y, τ) , where the normalized temporal coordinate is defined as

$$\tau = \frac{t}{T_{\max}}. \quad (47)$$

The architecture contains 4 hidden layers, 128 neurons per hidden layer, and $\tanh$ activation functions.

The output of the network is the scalar concentration field $u(x, y, t)$. The smoothness of the $\tanh$ activation function enables continuous evaluation of second-order derivatives required for the Laplacian operator.

7.4.2 Collocation Point Sampling

A total of 21,000 collocation points are employed using a structured multi-distribution sampling strategy with interior Points of $N_f = 15,000$, boundary Points of $N_b = 2,000$, and initial Condition of $N_i = 4,000$. Boundary points are further divided equally into early-time biases samples, uniform distributed temporal samples. This dual strategy prevents long-time mass retention.

Also, the initial-condition points are generated using a three-way distribution consisting of uniform sampling across the domain, tight clustering near 1σ , and broader sampling within 3σ . This strategy preserves the sharp gradients of the initial Gaussian pulse.

7.4.3 Loss Function Balancing

The total Mean Squared Error (MSE) loss function applies weighted penalties to balance the PDE residual, boundary conditions, and initial condition: $w_{\text{PDE}} = 1.0$, $w_{\text{BC}} = 50.0$, and $w_{\text{IC}} = 200.0$. The strong weighting of the initial condition prevents the PDE residual from smoothing away the localized Gaussian peak.

7.4.4 Multi-Phase Optimization

Adam Optimization The Adam optimizer is executed for 15,000 epochs using an initial learning rate of 10^{-3} which is gradually reduced to 10^{-5} through cosine annealing.

L-BFGS Refinement Following Adam optimization, 50 rounds of the quasi-Newton L-BFGS optimizer with strong Wolfe line search are applied.

This refinement phase reduces the residual loss to approximately 6.3×10^{-5} .

7.5 Results and Comparative Analysis

7.5.1 Spatio-Temporal Evolution

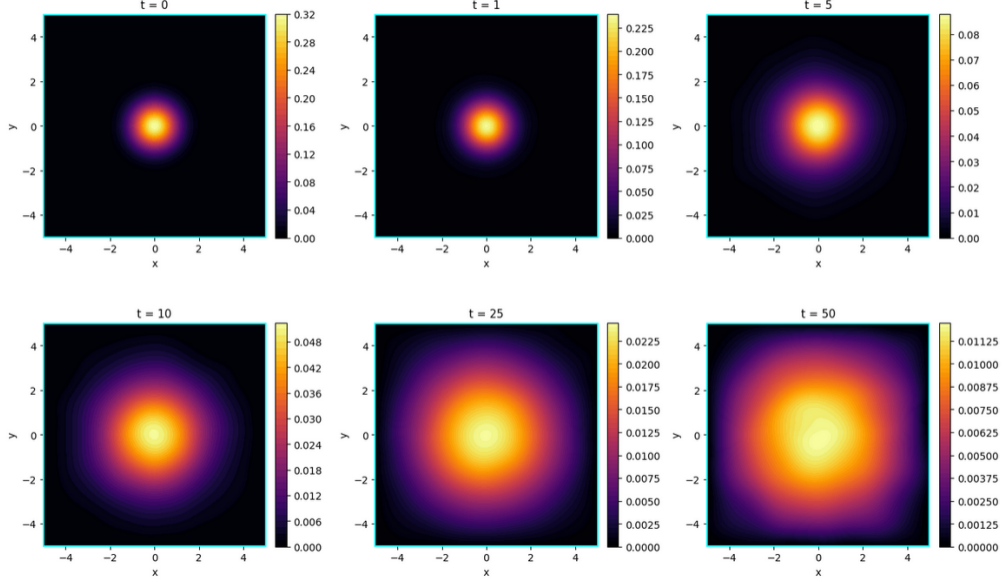


Figure 18: Solution snapshots demonstrating the two-dimensional diffusion process over time.

Both the PINN and Explicit FDM reproduce the expected diffusion dynamics.

Initially, the tightly localized Gaussian pulse spreads radially outward during the early stages of evolution.

As the concentration field reaches the absorbing boundaries, the originally circular diffusion gradually transitions into a square-shaped structure imposed by the domain geometry.

Continuous absorption at the four boundaries progressively reduces the overall concentration amplitude.

7.5.2 Transport Classification via Mean Squared Displacement

To classify the transport regime, the Mean Squared Displacement is computed and analyzed in log-log space during the early-time regime before strong boundary absorption dominates.

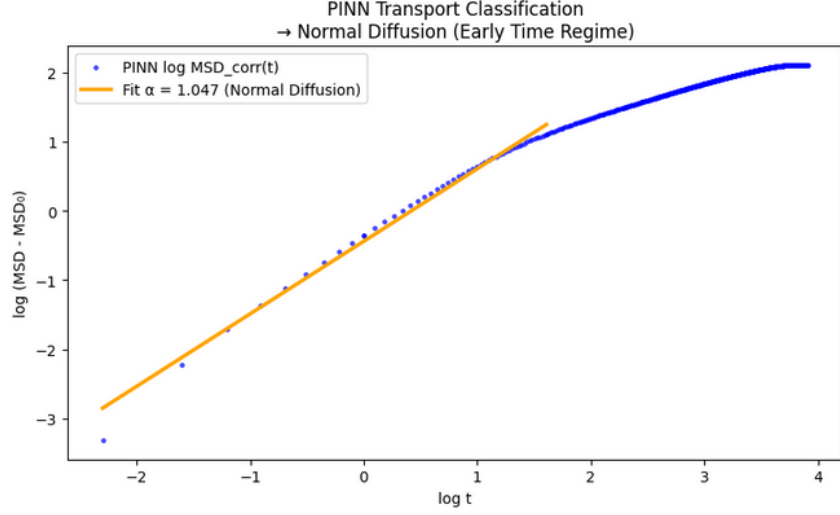


Figure 19: PINN Mean Squared Displacement regression analysis.

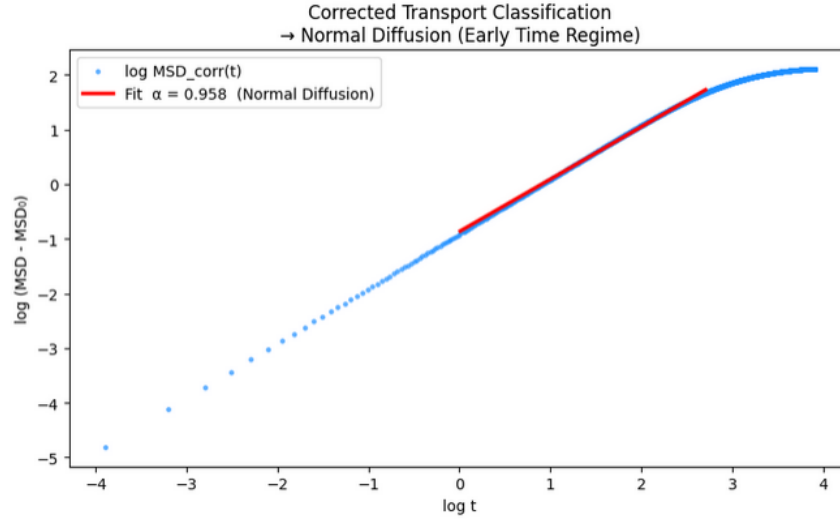


Figure 20: Explicit FDM Mean Squared Displacement regression analysis.

The extracted scaling exponents are: $\alpha_{\text{FDM}} = 0.958$, and $\alpha_{\text{PINN}} = 1.047$.

For classical normal diffusion, statistical mechanics predicts:

$$MSD(t) \propto t^\alpha, \quad \alpha = 1. \quad (48)$$

Both the Explicit FDM and the PINN therefore correctly classify the system within the normal diffusion regime.

7.5.3 Late-Time Survival Probability Decay

For a bounded square domain with absorbing boundaries, the late-time survival probability exhibits exponential decay governed by the lowest eigenvalue:

$$\lambda_1 = 2D \left(\frac{\pi}{L} \right)^2. \quad (49)$$

For $L = 10$ and $D = 0.1$,
the analytical decay slope becomes -0.01974 .

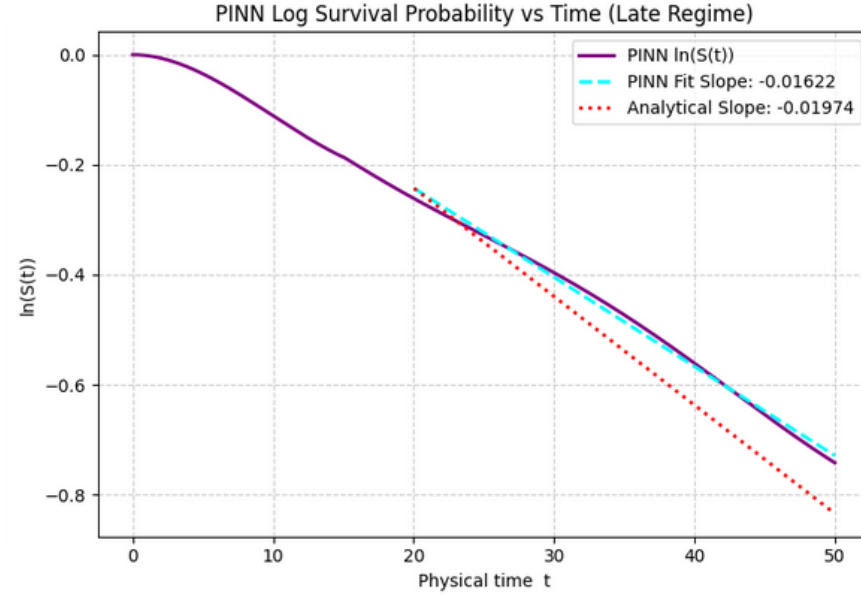


Figure 21: PINN late-time survival probability decay slope.

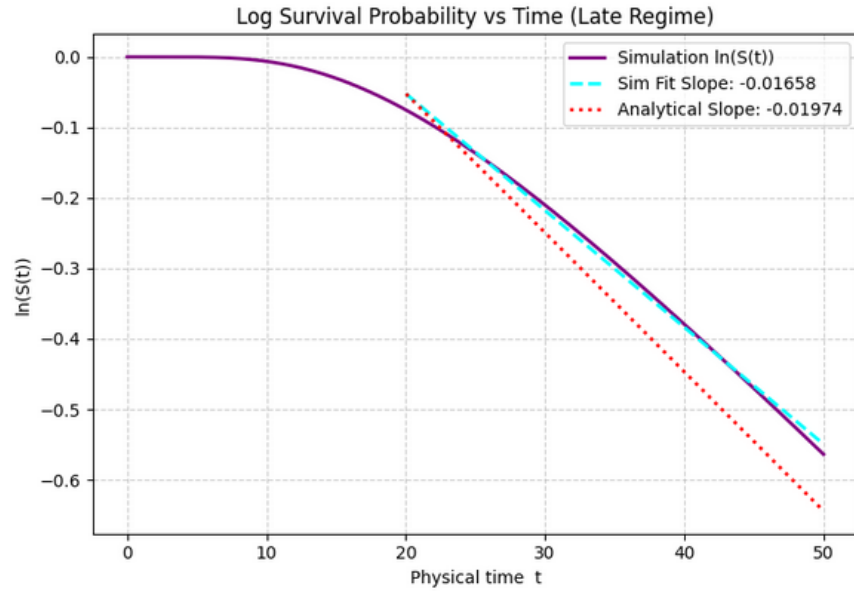


Figure 22: Explicit FDM late-time survival probability decay slope.

The extracted late-time slopes are:

$$\text{FDM slope} = -0.01658, \quad (50)$$

corresponding to an error of 16.02%.

The PINN produces:

$$\text{PINN slope} = -0.01622, \quad (51)$$

corresponding to an error of 17.81%.

7.6 Discussion

The bounded two-dimensional diffusion problem highlights the viability of the PINN as dependable framework for multidimensional transport phenomena.

The extension from one to two spatial dimensions forces the Explicit FDM to execute nearly 10,000 sequential time steps to satisfy CFL stability restrictions.

In contrast, the PINN approximates the complete global solution continuously across the full spatio-temporal domain.

The carefully designed network architecture, strong initial-condition weighting $w_{IC} = 200.0$, and dual-phase temporal sampling strategy successfully prevent neural-PDE failure.

The comparative transport metrics confirm that both approaches reproduce the expected statistical mechanics of normal diffusion.

8 The Inverse Problem: Discovering a Hidden Defect using PINNs

8.1 Abstract

Classical numerical methods are forward solvers. Meaning they need a specialised PDE with every physical parameters associated with it. But in practical cases, this critical parameters are often unknown and we must have to find it from sparse and noisy measurements.

This section investigates an inverse problem in which a 1-D rod containing a hidden physical defect acting as an unknown heat sink. While the Explicit FDM is used to generate synthetic sensor measurements, it cannot directly solve the inverse problem which marks a distinct advantage of PINN over FDM.

To address this challenge, a dual output PINN is constructed to:

1. Predict the temperature field,
2. Discover the hidden spatial defect profile purely from scattered noisy data.

8.2 Governing Equation and Data Generation

The physical system consists of a one-dimensional rod of length $L = 10.0$, governed by a modified diffusion equation containing an unknown sink term:

$$u_t = Du_{xx} - S(x)u. \quad (52)$$

The known physical parameters are the diffusion coefficient $D = 0.5$, initial Gaussian heat pulse centered at $x = 5.0$, with width $\sigma = 0.8$. The spatial boundaries satisfy homogeneous Dirichlet conditions: $u(0, t) = 0$, $u(10, t) = 0$.

8.2.1 Hidden Defect

A hidden defect of strength

$$S(x) = 2.0 \quad (53)$$

is artificially embedded within the interval $x \in [3.0, 4.5]$

The PINN is provided with no prior knowledge of this defect profile.

8.2.2 Sensor Data Generation

To generate synthetic measurement data, an Explicit FDM solver simulated the diffusion system over the temporal interval $T = 10.0$. From the dense numerical solution, exactly 2000 scattered sensor measurements were randomly sampled from the interior domain.

To create realistic experimental conditions, Gaussian measurement noise was added to the sensor data with approximate standard deviation: $\sigma \approx 0.00117$.

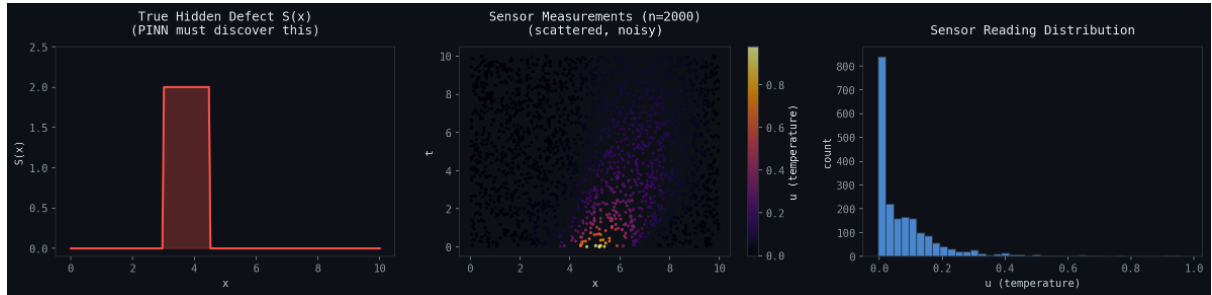


Figure 23: Phase 1 data overview. The left panel shows the hidden ground-truth defect profile, while the center and right panels display the scattered noisy sensor measurements provided to the PINN.

The PINN receives only the noisy sensor measurements and must infer the hidden physical defect entirely from the observed data.

8.3 PINN Architecture and Training Strategy

To solve the inverse problem, a specialized dual-network architecture named DualPINN was developed.

Rather than relying on a single neural network, the framework simultaneously optimizes two independent neural networks.

8.3.1 Network Architecture

Temperature Network The network

$$\text{net}_u, \quad (54)$$

takes the spatio-temporal coordinates (x, t) as inputs and predicts the dynamic temperature field $u(x, t)$. Its architecture consists of 6 hidden layers, 128 neurons per layer, Tanh activation functions.

Defect Network The second network,

$$\text{net}_S, \quad (55)$$

takes only the spatial coordinate x as input and predicts the unknown defect profile $S(x)$ as output. This network contains 4 hidden layers, 64 neurons per layer. The final output layer utilizes the `softplus` activation function to enforce the physical constraint, $S(x) \geq 0$. This guarantees that the defect behaves only as a sink and never as a heat source.

8.3.2 Loss Function Formulation

The loss function of this PINN involves three different types of loss. First one is the data loss which minimizes the mean square error between the PINN predictions and sensor data. The second one is the physics loss function itself, which enforces the PINN to validate the PDE $u_t = Du_{xx} - S(x)u$ across the entire domain. The third loss function is a regularization penalty which prevents the network from delusionalizing the defect strength across the entire domain.

8.3.3 Optimization Protocol

The DualPINN was trained for 20,000 epochs using the Adam optimizer. During the first 5000 epochs, the physics loss weight was reduced to: $\lambda_{\text{phys}} = 0.1$.

After this warm-up phase, the weight was increased to: $\lambda_{\text{phys}} = 1.0$.

This staged weighting prevents convergence towards solutions such as: $u = 0, S(x) = 0$. Which will make the PINN learn the noisy data first before enforcing the governing physics.

8.4 Results and Defect Discovery

The DualPINN successfully converged after approximately 558.4 seconds and have predicted both the temperature field and the hidden defect profile.

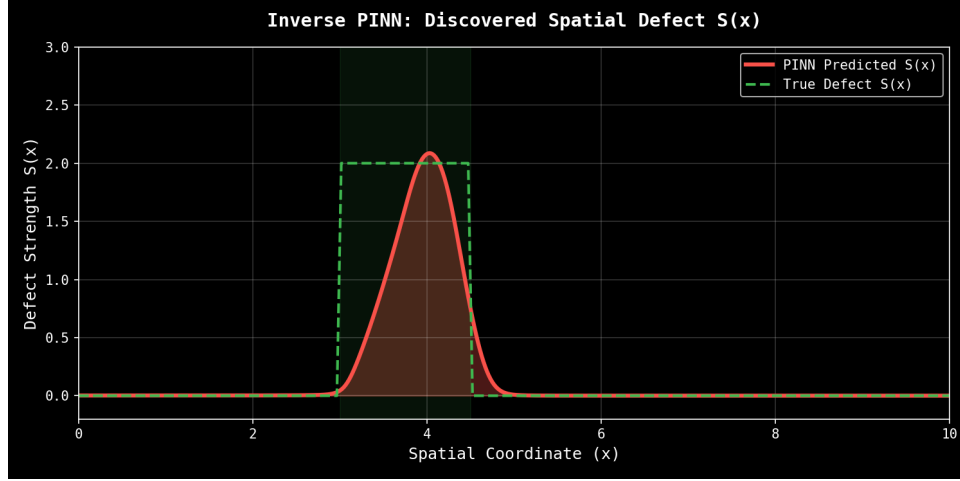


Figure 24: Comparison of the PINN predicted defect profile against the hidden true defect distribution.

The trained defect network `net_S` was evaluated across the domain to analyze its defect discovery capability.

Location Detection The network identified a strong defect region spanning: $x \in [3.07, 4.76]$, which closely matches the true hidden interval: $x \in [3.0, 4.5]$.

Intensity Estimation The PINN predicted a peak defect strength of: $S = 2.0861$ located near $x = 4.03$. This prediction deviates only slightly from the true hidden intensity: $S = 2.0$.

8.5 Discussion

This inverse problem marks one of the greatest advantage of PINN above FDM in scientific computing. Classical numerical methods like FDM are forward problem solvers which needs a specific PDE and the physical parameters of the system. This conditions are infact a constrain of FDM to use in practical situation. The physical system with all parameters known is a difficult problem in practical world. The practical problems always contain messy equations with several parameters unknown. PINN treats this parameters as additional trainable output. In this section the implemented PINN accurately reconstructed both the location and magnitude of the hidden defect, demonstrating the strong capability of PINNs for inverse problems, parameter discovery, and data driven scientific diagnosis.

9 Project Conclusion: Strategic Selection of Computational Solvers

9.1 Summary of the Computational Journey

Our project till now was a rigorous comparison of transport dynamics and statistical mechanics of the physical system using two different computational methods. This computational approaches for solving PDEs were:

1. The classical grid-based Explicit Finite Difference Method (FDM),
2. The modern mesh-free Physics-Informed Neural Network (PINN).

Throughout this project we had dealt with variety of physical systems involving 1-D domain, extending to 2-D domain, and an inverse problem involving discovery of parameter.

The conclusion of this project that both PINN and FDM are highly capable frameworks for solving PDEs. Rather than just seeing them as competing methods we should see them as complementary approaches with different strengths and limitations. A physicist solving this kind of computational problems must understand each methods strength and must use the method which is adequate for the specific problem in which he/she is working.

9.2 The First Key: FDM for Standard Forward Problems

Throughout this project whenever we are dealing with a forward problem which involves 1-D bounded diffusion problems to 2-D problems, Explicit FDM consistently showed an exceptional computational efficiency. While PINN involves a highly complex optimisation technique which involves navigating through a highly complex convex loss function, which in turn pay off with computational cost, FDM always produced accurate transport metrics with less computational cost. For a standard forward problem involving well defined geometries and manageable boundaries, FDM must be considered as the primary computational tool.

9.3 The Second Key: PINNs for Complex Boundaries

Even though, Explicit FDM is highly computationally efficient, it depends on spatial discretisation. When a physical system involving highly complex geometry and different boundaries like Neumann boundaries, FDM remains less efficient because of its discretization technique. In contrast, PINN operates as a mesh free framework which produce a continuous output function. Through automatic differentiation, the PINN can enforce Neumann and Dirichlet boundaries easily. Further its mesh free nature makes it capable of solving problems involving complex irregular geometries.

9.4 The Ultimate Key: PINNs for Inverse Problems

Inverse problem were the only problem in this project were the distinction between PINN and Explicit FDM became so obvious. While FDM and PINN has its own strengths and limitations, for dealing with an inverse problem PINN becomes the only method. PINNs capability to minimize the PDE residual simultaneously with data loss, made it a trustable tool to solve inverse problems which involves undefined physical parameter. In such problems PINN considers the undefined physical parameter as a trainable output and predicts the desired output. This capability is considered as one of the most transformative advantages of PINN in modern scientific computing.

9.5 Final Observations

Based on the comparative analysis conducted throughout this project, the following are the keen observations:

9.5.1 The Nature of Error

Both methods have different sources of numerical error.

Finite Difference Methods Explicit FDM method primarily suffer from cumulative truncation error. Small local approximation errors introduced at each time step gradually get accumulated over long simulations, slightly changing the late time transport dynamics such as decay exponents wrongly.

Physics-Informed Neural Networks PINNs inturn sufferes from optimization error. A perfect design of nueral architechtures, sampling strategies, pre-training phases, and balanced loss weighting are essentail for an accurate prediction. However, once properly converged, the PINN produces a global continuous function whose error does not accumulate progressively in time.

9.5.2 Front-Loaded vs Distributed Computational Cost

The two methods also exhibit fundamentally different computational cost distributions.

Explicit FDM The computational expense of FDM is distributed directly throughout the simulation execution. Each time step is inexpensive, but large numbers of iterations are required due to stability constraints introduced by the Courant–Friedrichs–Lewy condition.

PINNs In contrast, the computational cost of PINNs is almost entirely come from the training phase. The optimization process often requires thousands of epochs, multi stage optimizers, large collocation datasets etc.

However, once trained, the PINN can evaluate the solution instantaneously. But Explicit FDM remains to be the more computationally efficient framework in all projects we discussed so far.

9.5.3 Conclusion and Possible Future Extensions

The results of this project make us understand that deep learning framework is not a replacement for classical numerical solvers. Instead, the two methods must be complementary to each other. Explicit FDM remains the computationally efficient method in structured forward problems, but when the problem involves complex geometries, irregular boundaries, less observation data and inverse parameter discovery PINN becomes the capable tool. So one must only choose PINN whenever the system has high complex geometry where discretization is difficult, or when it is an inverse problem. In any other case classical numerical methods are advised to be the primary tool based on the observations from this project.

The future extensions of this project involve testing the efficiency of PINN in 3-dimensional systems, comparison between PINN and FDM in complex geometry such as a system which has absorbing geometries inside the absorbing boundary, and implementing PINN for highly irregular boundaries.

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